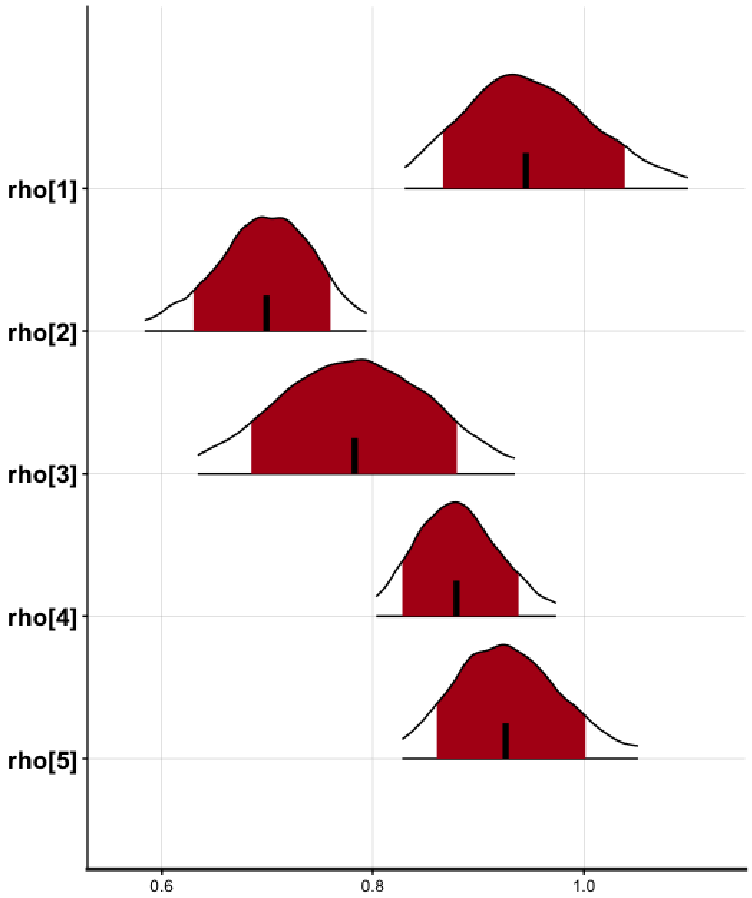
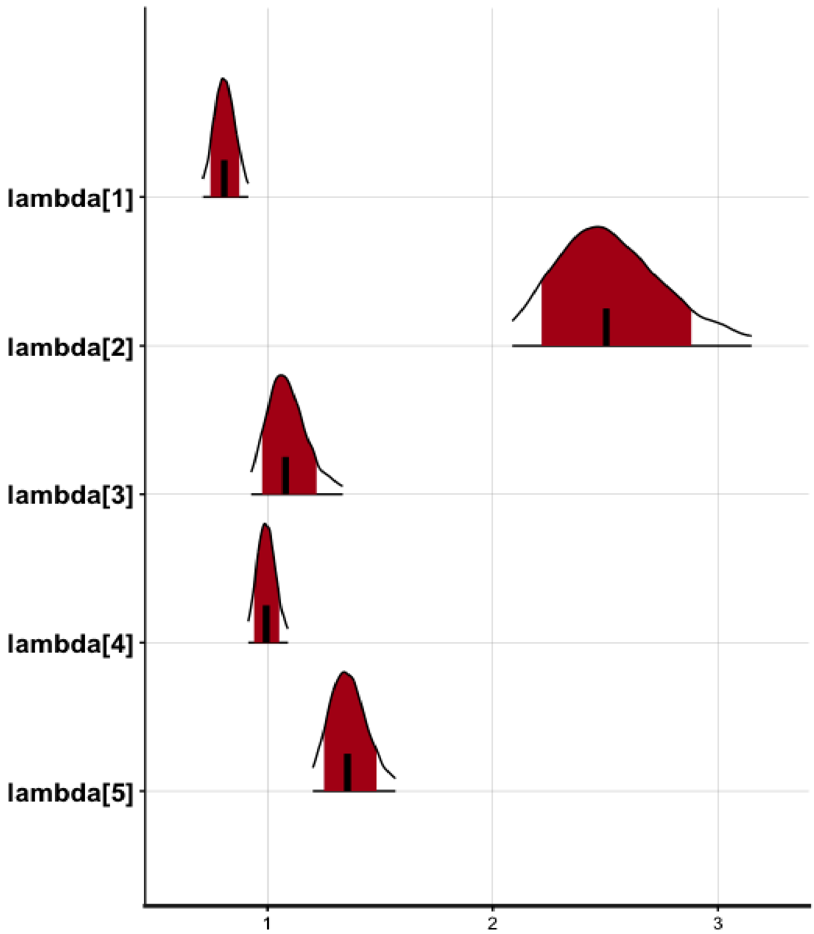
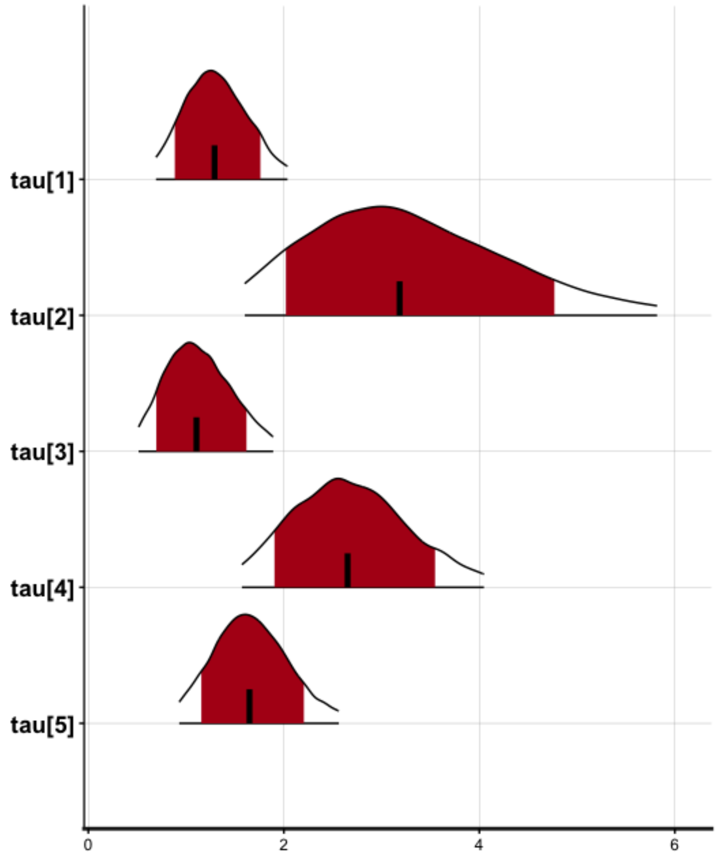
Q1 (1): Posterior distributions of the parameters of five subjects using ra_prospect model in hBayesDMM







(2) Reported posterior means and MLE estimates for subjects 2,3,4,6,7

---- Q1: hBayesDM posterior means ----

	subjID	rho	lambda	tau
1	2	0.9501318	0.8101103	1.310835
2	3	0.6968216	2.5334844	3.318409
3	4	0.7826464	1.0927184	1.133098
4	6	0.8814354	0.9962107	2.694073
5	7	0.9291778	1.3622193	1.671431

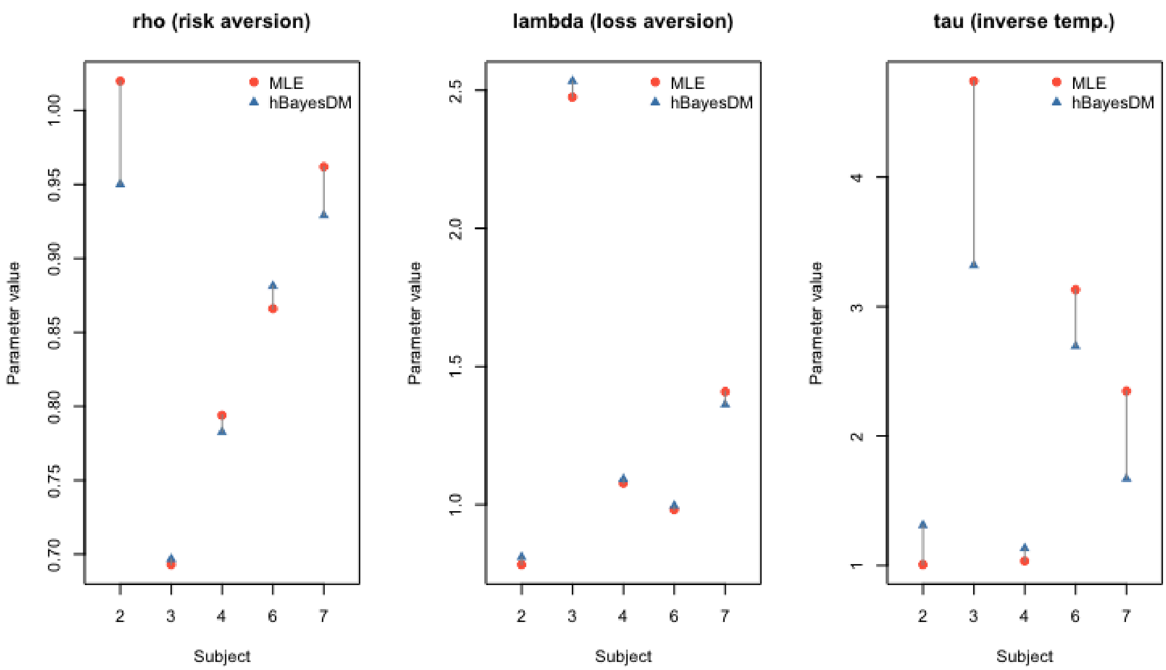
	subjID	rho_mle	lambda_mle	tau_mle
1	2	1.020	0.783	1.007
2	3	0.693	2.475	4.742
3	4	0.794	1.077	1.035
4	6	0.866	0.982	3.130
5	7	0.962	1.409	2.347

```
> print(output_hbayes$fit)
Inference for Stan model: anon_model.
4 chains, each with iter=4000; warmup=1000; thin=1;
post-warmup draws per chain=3000, total post-warmup draws=12000.

               mean se_mean   sd    2.5%     25%     50%     75%   97.5% n_eff Rhat
mu_rho         0.85    0.00 0.07    0.70    0.81    0.85    0.89    1.00  5218    1
mu_lambda      1.35    0.00 0.26    0.87    1.17    1.33    1.50    1.93  4403    1
mu_tau         1.99    0.01 0.58    1.10    1.60    1.91    2.27    3.36  5957    1
sigma[1]       0.18    0.00 0.07    0.07    0.13    0.17    0.22    0.35  6764    1
sigma[2]       0.34    0.00 0.09    0.20    0.28    0.33    0.40    0.55  8556    1
sigma[3]       0.26    0.00 0.10    0.08    0.19    0.25    0.32    0.48  6292    1
rho[1]         0.95    0.00 0.07    0.83    0.90    0.94    0.99    1.10 12893    1
rho[2]         0.70    0.00 0.05    0.58    0.67    0.70    0.73    0.79 13046    1
rho[3]         0.78    0.00 0.08    0.63    0.73    0.78    0.83    0.93 13632    1
rho[4]         0.88    0.00 0.04    0.80    0.85    0.88    0.91    0.97 13533    1
rho[5]         0.93    0.00 0.06    0.83    0.89    0.93    0.96    1.05 14084    1
lambda[1]      0.81    0.00 0.05    0.71    0.78    0.81    0.84    0.91 11649    1
lambda[2]      2.53    0.00 0.27    2.09    2.35    2.50    2.69    3.15 10106    1
lambda[3]      1.09    0.00 0.10    0.93    1.02    1.08    1.15    1.33  9814    1
lambda[4]      1.00    0.00 0.04    0.91    0.97    0.99    1.02    1.09 11394    1
lambda[5]      1.36    0.00 0.09    1.20    1.30    1.35    1.42    1.57 12829    1
tau[1]         1.31    0.00 0.34    0.69    1.07    1.29    1.54    2.04 12752    1
tau[2]         3.32    0.01 1.09    1.60    2.53    3.18    3.97    5.82 10838    1
tau[3]         1.13    0.00 0.36    0.51    0.87    1.11    1.37    1.89 11726    1
tau[4]         2.69    0.01 0.64    1.57    2.24    2.65    3.09    4.05 13209    1
tau[5]         1.67    0.00 0.41    0.93    1.39    1.65    1.93    2.56 16135    1
log_lik[1]   -70.42    0.01 1.34  -73.78  -71.06  -70.10  -69.42  -68.82  7976    1
log_lik[2]   -20.65    0.02 1.56  -24.40  -21.49  -20.32  -19.48  -18.66  6056    1
log_lik[3]   -82.34    0.01 1.09  -85.13  -82.84  -82.07  -81.54  -81.11  6273    1
log_lik[4]   -45.46    0.02 1.34  -48.89  -46.09  -45.12  -44.48  -43.91  7656    1
log_lik[5]   -47.04    0.01 1.16  -50.08  -47.56  -46.73  -46.19  -45.73  7370    1
lp__        -283.64    0.07 3.88 -292.23 -286.02 -283.29 -280.85 -277.15  3247    1
```

Convergence check for parameters: we can see that the Rhat values are close to 1.

(3) Visual comparison of MLE estimates and posterior means



Using hBayesDM and maximum likelihood estimation (MLE) can lead to different results. For MLE, the estimation is drawn independently from the subject's own trials. However, a hierarchical model is used for hBayesDM, wherein the parameters are drawn from a group-level distribution. Shrinkage also occurs in hBayesDM, as single-subject estimates are pulled towards the group mean. It is also important to note while MLE is a single point, hBayesDM returns a distribution. Hence, comparing the mean of the distribution and MLE results can be different, as MLE instead shows where the likelihood peaks.

Q2 (1) : The Bayes Factor (BF) of model 1 and 2 = 3.245351e+15

(2): According to Jeffrey's evidence categories, a BF value of 3.245351e+15 shows that we have decisive evidence for model 1 (ra_prospect) compared to model 2 (ra_noLA), as it is greater than 100.

(3) The posterior odds are 4.868027e+15.

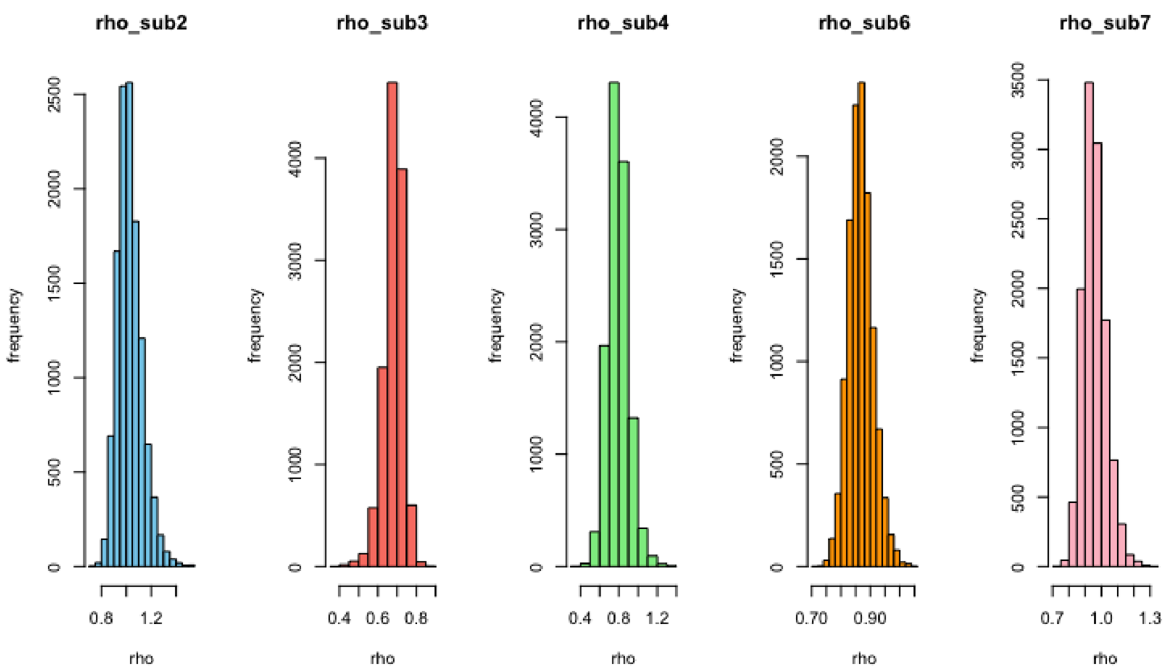
(4) The posterior model probability is 1.

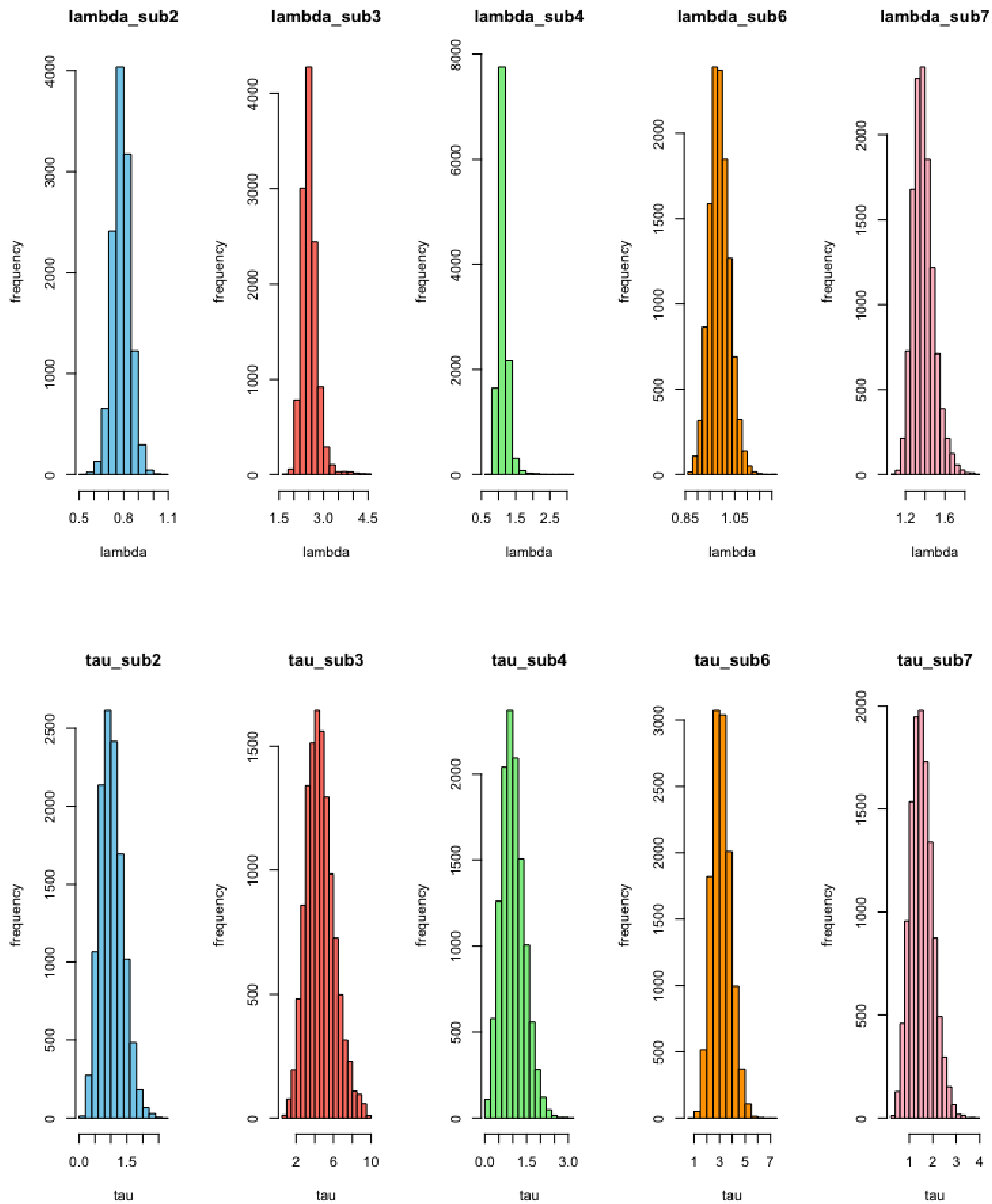
(5) Bayes Factor (BF) for seven memory retention models

BF was compared against the best model, POW1, with a BIC value of 475.292.

	model	BIC	BF_best_vs_model	postProb
1	POW2	477.371	2.828000e+00	0.2292
2	EXP2	489.335	1.120466e+03	0.0006
3	POW1	475.292	1.000000e+00	0.6480
4	EXP1	524.532	4.924137e+10	0.0000
5	EXPOW	479.451	8.000000e+00	0.0810
6	HYP1	482.954	4.610900e+01	0.0141
7	HYP2	481.634	2.383100e+01	0.0272

Q3 (1): Posterior distribution of all 5 subjects using Stan





(2) Posterior means of 5 subjects

	RHO	LAMBDA	TAU
subject2	1.0287357	0.7853898	1.019963
subject3	0.6811066	2.5260091	4.590190
subject4	0.7928410	1.1242137	1.005956
subject6	0.8673410	0.9881507	3.133194
subject7	0.9572640	1.3822399	1.529103

	mean	se_mean	sd	2.5%	25%	50%	75%	97.5%	n_eff	Rhat
rho[1]	1.03	0.00	0.10	0.87	0.96	1.02	1.09	1.26	5661	1
rho[2]	0.68	0.00	0.05	0.56	0.65	0.69	0.71	0.76	7714	1
rho[3]	0.79	0.00	0.11	0.60	0.72	0.78	0.86	1.03	6764	1
rho[4]	0.87	0.00	0.04	0.79	0.84	0.87	0.89	0.96	9879	1
rho[5]	0.96	0.00	0.07	0.84	0.91	0.95	1.00	1.12	6632	1
lambda[1]	0.79	0.00	0.06	0.67	0.75	0.78	0.82	0.90	10468	1
lambda[2]	2.53	0.00	0.27	2.10	2.36	2.50	2.66	3.10	3846	1
lambda[3]	1.12	0.00	0.15	0.93	1.04	1.10	1.18	1.46	3350	1
lambda[4]	0.99	0.00	0.04	0.91	0.96	0.99	1.01	1.07	13954	1
lambda[5]	1.38	0.00	0.11	1.21	1.31	1.37	1.44	1.63	8265	1
tau[1]	1.01	0.00	0.36	0.39	0.74	0.98	1.24	1.80	7719	1
tau[2]	4.57	0.02	1.48	2.05	3.51	4.43	5.48	7.83	7298	1
tau[3]	1.01	0.00	0.41	0.30	0.71	0.97	1.26	1.89	7317	1
tau[4]	3.12	0.01	0.74	1.79	2.62	3.08	3.59	4.70	10408	1
tau[5]	1.52	0.01	0.48	0.68	1.18	1.48	1.83	2.56	7726	1
lp__	-268.28	0.05	3.03	-275.28	-270.05	-267.91	-266.14	-263.42	3871	1

Convergence check: we can see that Rhat values are close to 1.